EMC TEST REPORT

Applicant	:	Shenzhen HOPE Microelectronics Co., Ltd.
Address of Applicant	:	30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district, Nanshan, Shenzhen, GD, P.R. China
Manufacturer	:	Shenzhen HOPE Microelectronics Co., Ltd.
Address of Manufacturer	:	30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district, Nanshan, Shenzhen, GD, P.R. China
Equipment under Test	:	LoRa module
Model No.	:	RFM90C
Test Standard(s)	:	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-3 V2.3.2 (2023-01)
Report No.	:	DDT-RE25091517-2E01
Issue Date	:	2025/12/25
Issued By	:	Guangdong Dongdian Testing Service Co., Ltd. Unit 2, Building 1, No. 17, Zongbu 2nd Road, Songshan Lake Park, Dongguan, Guangdong, China, 523808

REPORT

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Test Report Declare

Applicant	:	Shenzhen HOPE Microelectronics Co., Ltd
Address	:	30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China
Manufacturer	:	Shenzhen HOPE Microelectronics Co., Ltd
Address	:	30/F, Block A, Building 8, Phase 3,Vanke Cloud City,Xili Sub-district,Nanshan,Shenzhen,GD,P.R.China
Equipment under Test	:	LoRa module
Model No.	:	RFM90C

Test Standard Used:
ETSI EN 301 489-1 V2.2.3 (2019-11), ETSI EN 301 489-3 V2.3.2 (2023-01)

We Declare:
The equipment described above is tested by Guangdong Dongdian Testing Service Co., Ltd. and in the configuration tested the equipment complied with the standards specified above. The test results are contained in this test report and Guangdong Dongdian Testing Service Co., Ltd. is assumed of full responsibility for the accuracy and completeness of these tests.

Report No.:	DDT-RE25091517-2E01		
Date of Receipt:	2025/12/03	**Date of Test:**	2025/12/03-2025/12/05

Created: David Gao	Reviewed: Caesar Peng	Approved: Damon Hu
David Gao	Caesar Peng	Damon Hu
2025/12/05	2025/12/25	2025/12/25

Note: This report applies to above tested sample only. This report shall not be reproduced in parts without written approval of Guangdong Dongdian Testing Service Co., Ltd.

Revision History

Version	Revision Content	Issue Date	Approved
V0	Initial issue	2025/12/25	Damon Hu

1. Summary of Test Results

EMISSION (EMI)			
Description of Test Item	**Standard**	**Result**	**Memo**
DC Power Port Conducted Emission	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Vehicle Use
AC Power Port Conducted Emission	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Only for AC power port
Wired network port Conducted Emission	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	For signal port
Radiated Emissions Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**PASS**	/
Harmonic Current Emissions Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	For AC power port (≤16A), According to IEC 61000-3-2 Clause 7, this product belongs to exceptions of Clause 7 or Annex C. limits are not specified in this standard.
Voltage Fluctuations& Flicker Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	AC power port (≤16A)
IMMUNITY (EMS)			
Description of Test Item	**Standard**	**Result**	**Memo**
Electrostatic Discharge Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**PASS**	/
Continuous Radio Frequency Disturbances Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**PASS**	/
Electrical Fast Transients(EFT) Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Applicable to AC power port. For analogue/digital data ports and DC network power ports, applicable only to ports which, according to the manufacturer's specification, support cable lengths greater than 3 m.
Surge Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Applicable to AC power port. For analogue/digital data ports and DC network

			power ports, applicable only to ports which, according to the manufacturer's specification, support cable lengths greater than 3 m.
Continuous Conducted Disturbances Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Applicable to AC power port. For analogue/digital data ports and DC network power ports, applicable only to ports which, according to the manufacturer's specification, support cable lengths greater than 3 m.
Voltage Dips and Interruptions Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Only for AC power port
Transients and Surge for DC Port Test	ETSI EN 301 489-1 V2.2.3 (2019-11),ETSI EN 301 489-3 V2.3.2 (2023-01)	**N/A**	Vehicle Use

Note: N/A is an abbreviation for Not Applicable, and means this item is not applicable for this device or no need to test according to standard.

Note: 1. The EMI measurements had been made in the operating mode producing the largest emission in the frequency band being investigated consistent with normal applications.
An attempt had been made to maximize the emission by varying the configuration of the EUT.

2. The EMS measurements had been made in the frequency bands being investigated, with the EUT in the most susceptible operating mode consistent with normal applications. The configuration of the test sample had been varied to achieve maximum susceptibility.

Remark: Performance criteria:

Criteria	During test	After test (i.e. as a result of the application of the test)
A	Shall operate as intended. (see note). Shall be no loss of function. Shall be no unintentional transmissions.	Shall operate as intended. Shall be no degradation of performance. Shall be no loss of function. Shall be no loss of critical stored data.
B	May be loss of function.	Functions shall be self-recoverable. Shall operate as intended after recovering. Shall be no loss of critical stored data.
C	May be loss of function.	Functions shall be recoverable by the operator. Shall operate as intended after recovering. Shall be no loss of critical stored data.
NOTE: Operate as intended during the test allows a level of degradation in accordance with Minimum performance level. For equipment that supports a PER or FER, the minimum performance level shall be a PER or FER less than or equal to 10 %. For equipment that does not support a PER or a FER, the minimum performance level shall be no loss of the wireless transmission function needed for the intended use of the equipment.		

Performance criteria for continuous phenomena:

The performance criteria A shall apply.

During the test, the equipment shall:

• continue to operate as intended;

• not unintentionally transmit;

• not unintentionally change its operating state;

• not unintentionally change critical stored data.

Performance criteria for Transient phenomena:

The performance criteria B shall apply.

For all ports and transient phenomena with the exception described below, the following applies:

• The application of the transient phenomena shall not result in a change of the mode of operation (e.g. unintended transmission) or the loss of critical stored data.

• After application of the transient phenomena, the equipment shall operate as intended.

For surges applied to symmetrically operated wired network ports intended to be connected directly to outdoor lines the following criteria applies:

• For products with only one symmetrical port intended for connection to outdoor lines, loss of function is allowed, provided the function is self-recoverable, or can be otherwise restored. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

• For products with more than one symmetrical port intended for connection to outdoor lines, loss of function on the port under test is allowed, provided the function is self-recoverable. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

2. General Test Information

2.1. Description of EUT

EUT* Name	:	LoRa module
Model Number	:	RFM90C
Difference of model number	:	/
EUT Function Description	:	Please reference user manual of this device
Power Supply	:	DC 1.8~3.7V
Hardware Version	:	V1.2
Software Version	:	V1.0
Operation Frequency	:	868.1MHz, 868.3MHz, 868.5MHz, 864.1MHz, 864.3MHz,864.5MHz
Modulation	:	CSS
Antenna	:	SMA antenna, maximum PK gain: 3.0 dBi
Sample Number	:	S25091517-003

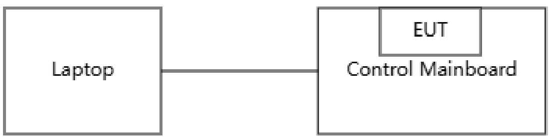
Note 1: EUT is the abbreviation of equipment under test.

2.2. Accessories of EUT

Accessories	Manufacturer	Model number	Description
/	/	/	/

2.3. Block diagram EUT configuration for test

Mode 1:Working mode



2.4. Decision of final test mode

According pre-test, the worst test modes were reported as below.

Emission	Radiated Emissions Test	Mode 1: Working mode
Immunity	Electrostatic Discharge Test	Mode 1: Working mode
	Continuous Radio Frequency Disturbances Test	Mode 1: Working mode

2.5. Deviations of test standard

No deviation.

2.6. Test environment conditions

During the measurement the environmental conditions were within the listed ranges:

Temperature range:	20-25℃
Humidity range:	40-75%
Pressure range:	86-106 kPa

Note: The specific temperature and humidity information of each test item refers to the temperature and humidity record in the corresponding test data.

2.7. Test laboratory

Guangdong Dongdian Testing Service Co., Ltd.

Add.: Unit 2, Building 1, No. 17, Zongbu 2nd Road, Songshan Lake Park, Dongguan, Guangdong, China, 523808.

Tel.: +86-0769-38826678, http://www.dgddt.com, Email: ddt@dgddt.com.

CNAS Accreditation No. L6451; A2LA Accreditation Number: 3870.01

FCC Designation Number: CN1182, Test Firm Registration Number: 540522

Innovation, Science and Economic Development Canada Site Registration Number: 10288A

Conformity Assessment Body identifier: CN0048

VCCI facility registration number: C-20087, T-20088, R-20123, R-20240, G-20118

2.8. Measurement uncertainty

Test Item	Uncertainty
Conducted disturbance at mains terminals	1#: 3.72dB (9 kHz to 150 kHz), 3.34dB (150 kHz to 30 MHz)
	2#: 3.75dB (9 kHz to 150 kHz), 3.39dB (150 kHz to 30 MHz)
	3#: 3.78dB (9 kHz to 150 kHz), 3.37dB (150 kHz to 30 MHz)
Uncertainty for telecommunication port conduction emission test	1#: AAN with aLCL = 55 ... 40 dBc: 3.64 dB AAN with aLCL = 65 ... 50 dBc: 4.08 dB AAN with aLCL = 75 ... 60 dBc: 4.56 dB
	2#: AAN with aLCL = 55 ... 40 dBc: 3.82 dB AAN with aLCL = 65 ... 50 dBc: 3.96 dB AAN with aLCL = 75 ... 60 dBc: 4.12 dB
Uncertainty for radiation emission test (30 MHz-1 GHz)	1#: 4.94 dB (Antenna Polarize: V) 4.68 dB (Antenna Polarize: H)
	2#: 4.94 dB (Antenna Polarize: V) 4.68 dB (Antenna Polarize: H)
	3#: 4.96 dB (Antenna Polarize: V) 4.98 dB (Antenna Polarize: H)
	10m: 4.48 dB (Antenna Polarize: V) 4.64 dB (Antenna Polarize: H)
Uncertainty for radiation disturbance test (1 GHz to 6 GHz)	1#: 4.10 dB (1-6 GHz)
	3#: 4.54 dB (1-6 GHz)
Uncertainty for Flicker test	0.2%
Uncertainty for Harmonic test	5%
Uncertainty for Electrostatic discharge	Rise time: 4% Peak current: 3.1% Current at 30 ns: 3.1% Current at 60 ns: 3.1%
Uncertainty for Surge	Peak of the open-circuit voltage impulse: 3% Front time of the open-circuit voltage impulse: 5% Width of the open-circuit voltage impulse: 5% Peak of the short-circuit current impulse: 2.7% Front time of the short-circuit current impulse: 5% Duration of the short-circuit current impulse: 3%
Uncertainty for Electrical fast transients	Voltage rise time: 3.7% Peak voltage value: 3.4% Voltage pulse width: 3.7%
Uncertainty for Continuous conducted disturbances	0.25dB
Uncertainty for Continuous radio frequency disturbances	1.12dB
Uncertainty for Power-frequency magnetic fields	10%
Uncertainty for Voltage dips and interruptions	3.7%
Temperature	0.4 ℃
Humidity	2%
Note: This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.	

3. Radiated Emissions Test

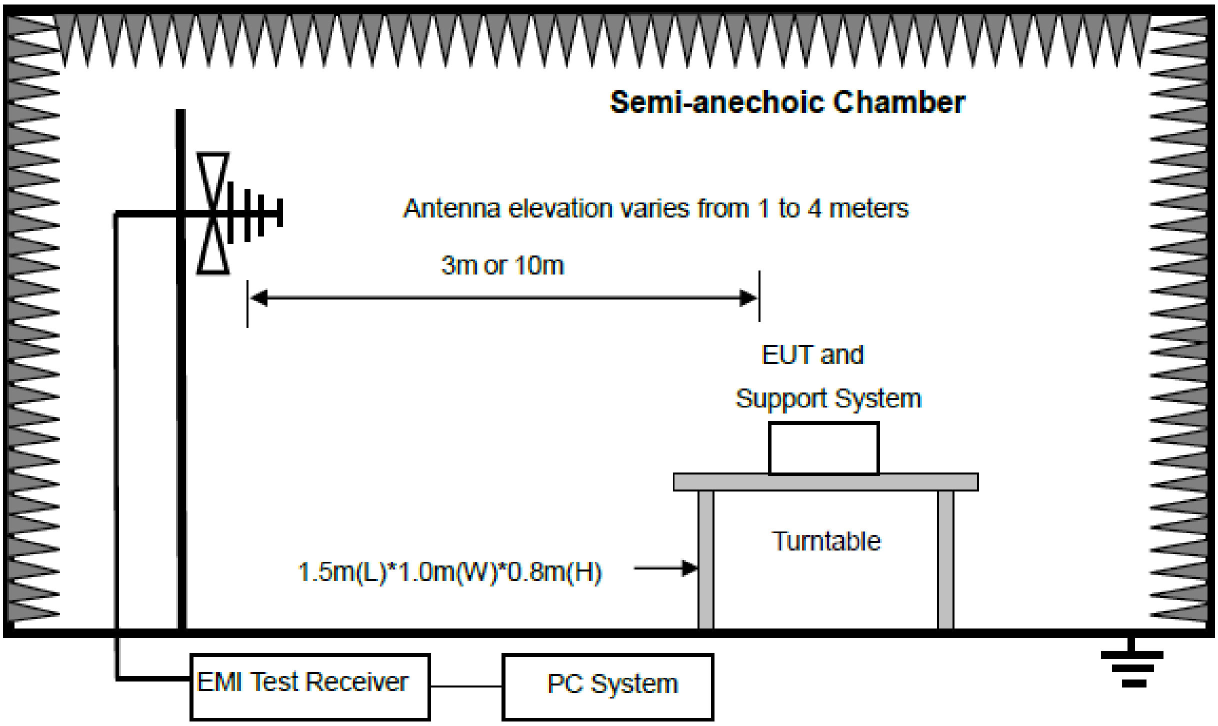
3.1. Test equipment

Equipment	Manufacturer	Model No.	Equipment No.	Cal Due To
Trilog Broadband Antenna	Schwarzbeck	VULB 9163	DDT-ZC00522	2026/07/22
RE Cable below 1GHz for 1#RE	R&S	ESU8/RF1	DDT-ZC00567	2026/07/06
Active Loop Antenna	Schwarzbeck	FMZB1519	DDT-ZC00524	2026/08/18
EMI Test Receiver	R&S	ESCI	DDT-ZC01297	2026/07/06
Spectrum Analyzer	Agilent	E4440A	DDT-ZC01445	2026/03/28
Broad-Band Horn Antenna	Schwarzbeck	BBHA 9170	DDT-ZC00506	2026/04/01
Horn Antenna	SCHWARZBECK	BBHA9120 D	DDT-ZC01218	2026/08/13
Preamplifier	COM-POWER	PAM-118A	DDT-ZC01489	2026/08/10
Pre-amplifier	COM-POWER	PAM-840A	DDT-ZC01693	2026/03/28
EMI Test Software	Audix	e3	DDT-ZC01252	/
RF cable	Zhongke Junchuang	JCTB810-NJ-NJ-7M	DDT-ZC02759	2026/07/22

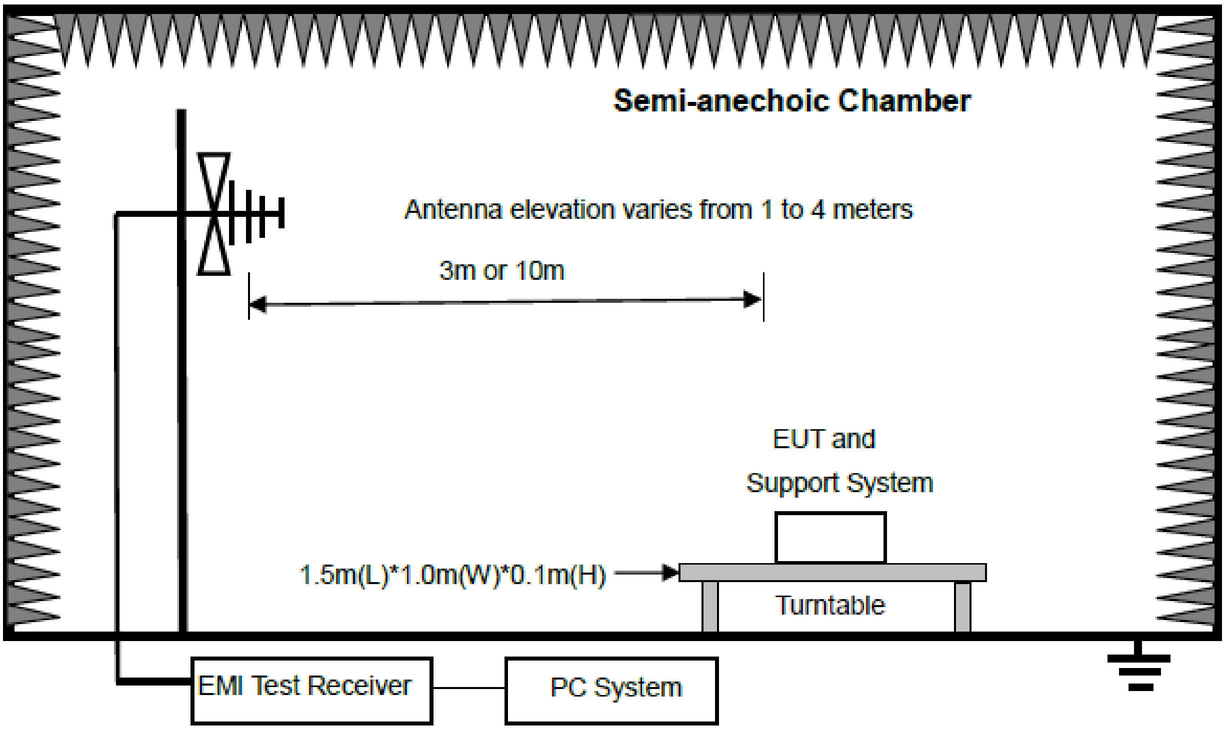
3.2. Block diagram of test setup

Below 1 GHz

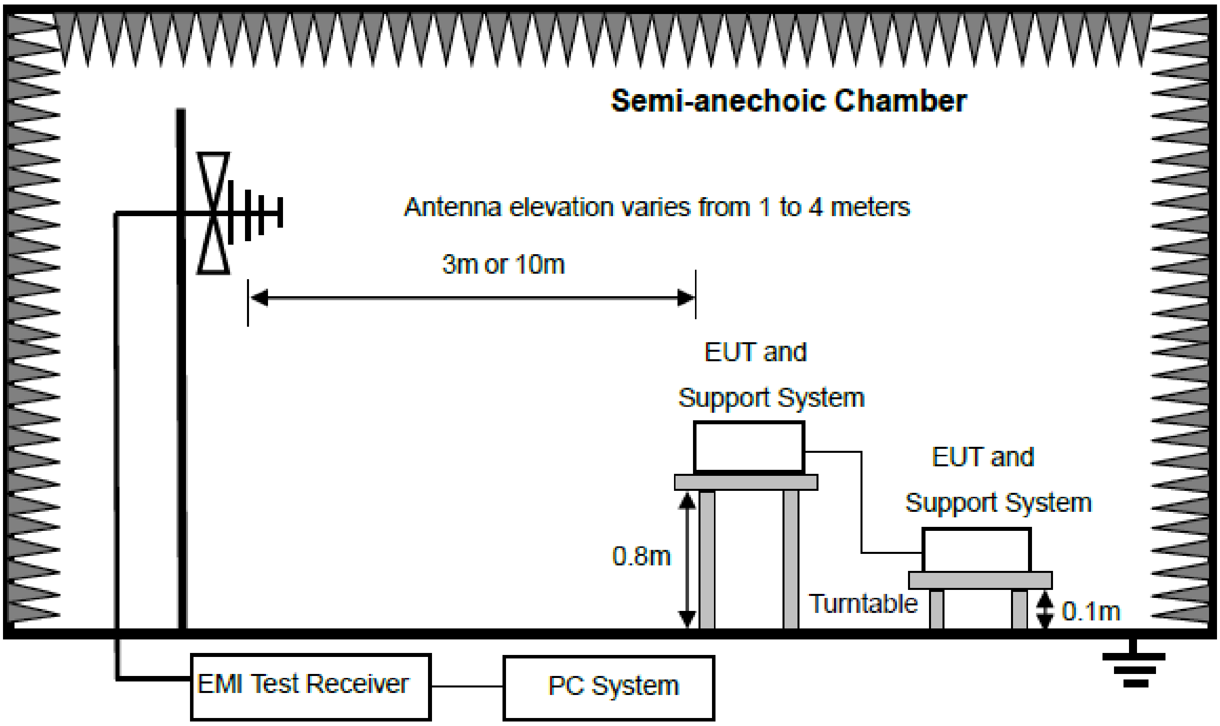
For table-top equipment



For floor standing equipment

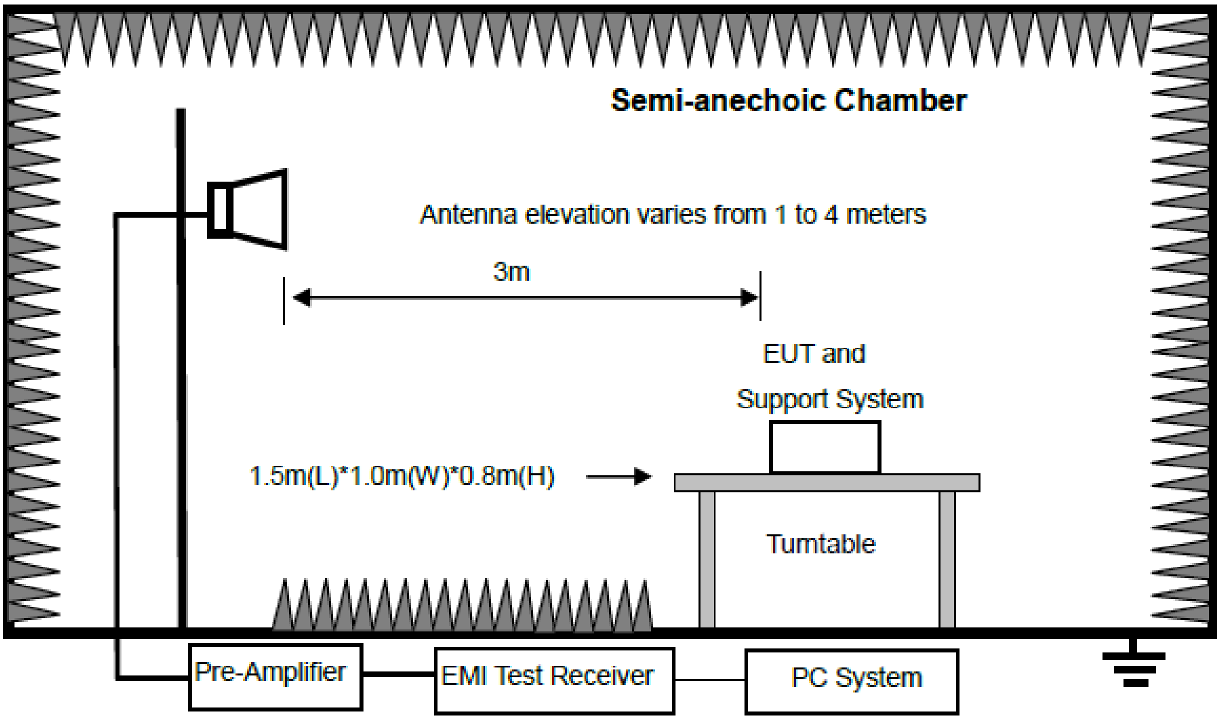


For combinations equipment

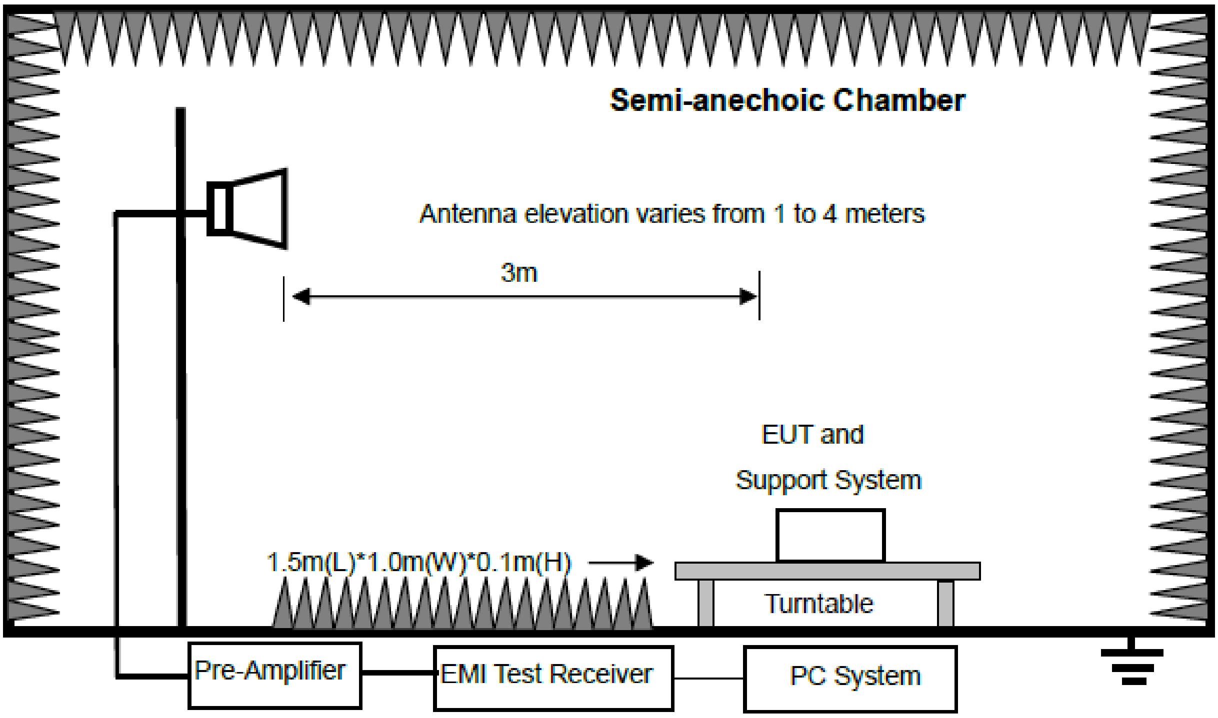


Above 1 GHz

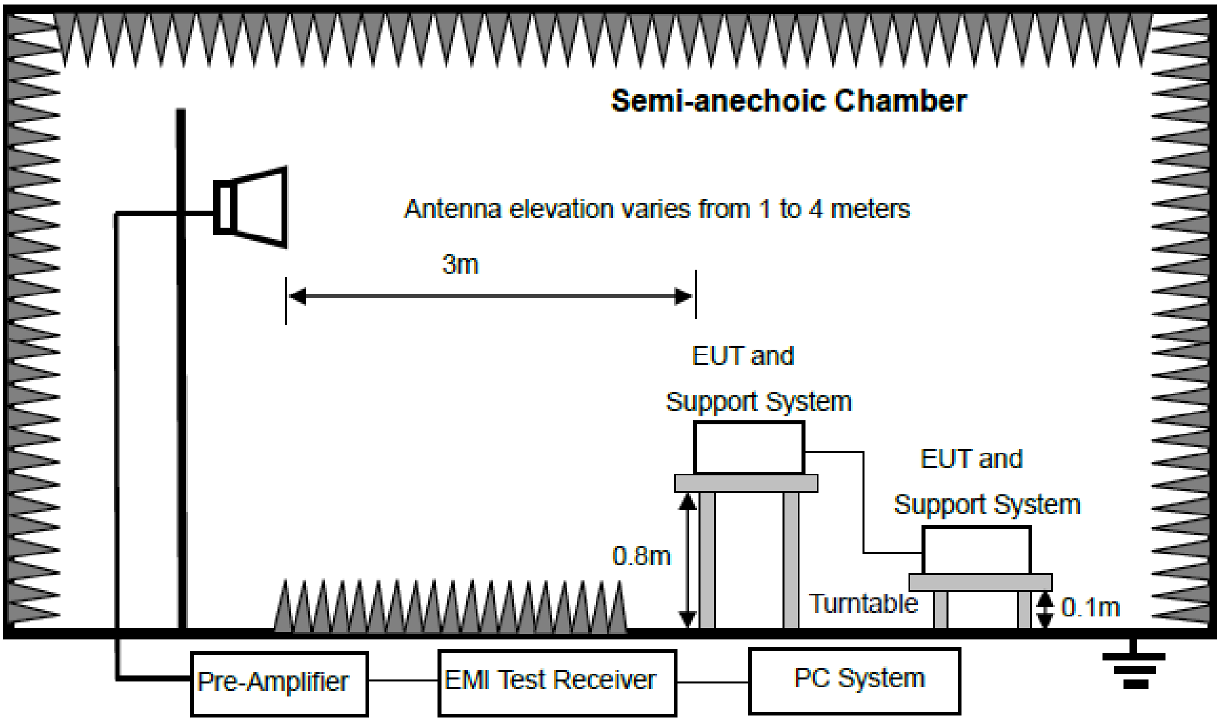
For table-top equipment



For floor standing equipment



For combinations equipment



3.3. Limits

Equipment	Frequency	Field Strengths Limits at 10m measuring distance dB(μV)/m	Field Strengths Limits at 3m measuring distance dB(μV)/m
Class A Equipment	30 MHz to 230 MHz	40	50
	230 MHz to 1000 MHz	47	57
	1 GHz to 3 GHz	/	Average:56; Peak:76
	3 GHz to 6 GHz	/	Average:60; Peak:80
Equipment	Frequency	Field Strengths Limits at 10m measuring distance dB(μV)/m	Field Strengths Limits at 3m measuring distance dB(μV)/m
Class B Equipment	30 MHz to 230 MHz	30	40
	230 MHz to 1000 MHz	37	47
	1 GHz to 3 GHz	/	Average:50; Peak:70
	3 GHz to 6 GHz	/	Average:54; Peak:74
Note: (1) The smaller limit shall apply at the cross point between two frequency bands. (2) Distance is the distance in meters between the measuring instrument, antenna and the closest point of any part of the device or system.			

3.4. Assistant equipment used for test

Assistant equipment	Manufacturer	Model number	Description	other
Laptop	HP	HP ProBook 445 G6	5CD9112VSV	/

3.5. Test procedure

(1) The EUT was placed on a non-metallic table (Refer to the 'Block diagram of test setup'). above the ground plane inside an anechoic chamber.
(2) Test antenna was located 3m (see note) from the EUT on an adjustable mast. A pre-scan was first performed in order to find prominent radiated emissions. For final emissions measurements at each frequency of interest, the EUT were rotated and the antenna height was varied between 1m and 4m in order to maximize the emission. Measurements in both horizontal and vertical polarities were made and the data was recorded. In order to find the maximum emission, the relative positions of equipment and all of the interface cables were changed according to ETSI EN 301 489 on radiated emission test.
(3) Spectrum frequency from 30 MHz to ☐1 GHz /☒6 GHz was investigated.
(4) For final emissions measurements at each frequency of interest, the EUT were rotated and the antenna height was varied between 1m and 4m in order to maximize the emission. Measurements in both horizontal and vertical polarities were made and the data was recorded. In order to find the maximum emission, the relative positions of equipment and all of the interface cables were changed according to ETSI EN 301 489 on Radiated Emission test.
(5) For emissions from 30 MHz to 1GHz, Quasi-Peak values were measured with EMI Receiver and the bandwidth of Receiver is 120 kHz.
(6) For emissions above 1 GHz, both Peak and Average level were measured with Spectrum Analyzer, and the RBW is set at 1 MHz VBW is set at 3 MHz.

3.6. Test result

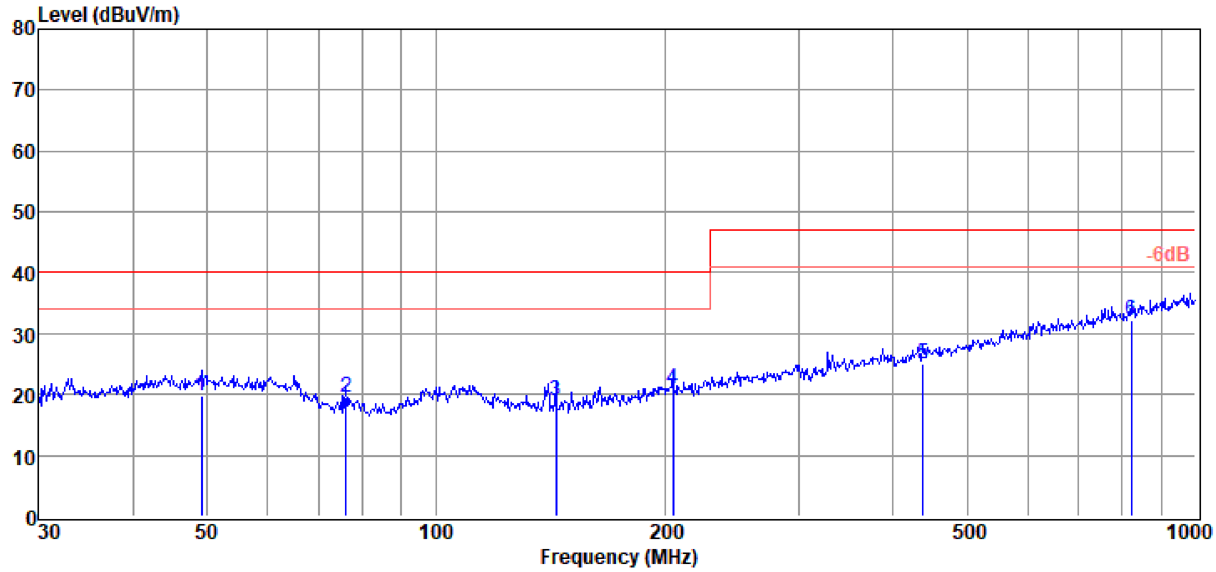
Pass. (See below detailed test result)
Note 1: All emissions not reported below are too low against the prescribed limits.
Note 2: “-----” means Peak detection; “-----” means Average detection.

3.7. Test data

TR-4-E-009 Radiated Emission Test Result

Test Site	: DDT 3m Chamber 1#	D:\2025 RE 1# Report data\Q25091517-2E\1203 RE.EM6	
Test Date	: 2025-12-03	Tested By	: Li guilin
Power Supply	: DC 3.3V from control board	Test Mode	: Working mode
Condition	: Temp:23.5°C,Humi:65.4%	Antenna/Distance	: 2025 9163 1#/3m/VERTICAL
Memo	: S25091517-003		

Data: 1



Item (Mark)	Freq. (MHz)	Read Level (dBµV)	Antenna Factor (dB/m)	Cable Loss dB	Result Level (dBµV/m)	Limit Line (dBµV/m)	Margin (dB)	Detector	Polarization
1	49.19	2.21	13.82	3.75	19.78	40.00	20.22	QP	VERTICAL
2	76.24	6.99	8.50	3.97	19.46	40.00	20.54	QP	VERTICAL
3	143.83	5.76	8.88	4.46	19.10	40.00	20.90	QP	VERTICAL
4	204.96	4.86	11.30	4.78	20.94	40.00	19.06	QP	VERTICAL
5	437.12	3.22	16.09	5.82	25.13	47.00	21.87	QP	VERTICAL
6	821.71	4.59	20.63	7.00	32.22	47.00	14.78	QP	VERTICAL

Note: 1. Result Level = Read Level + Antenna Factor + Cable loss.
2. If Peak Result complies with QP limit, QP Result is deemed to comply with QP limit.
3. Test setup: RBW: 120 kHz, VBW: 300 kHz, Sweep time: auto.

TR-4-E-009 Radiated Emission Test Result

Test Site : DDT 3m Chamber 1#

D:\2025 RE 1# Report data\Q25091517-2E\1203 RE.EM6

Test Date : 2025-12-03

Tested By : Li guilin

Power Supply : DC 3.3V from control board

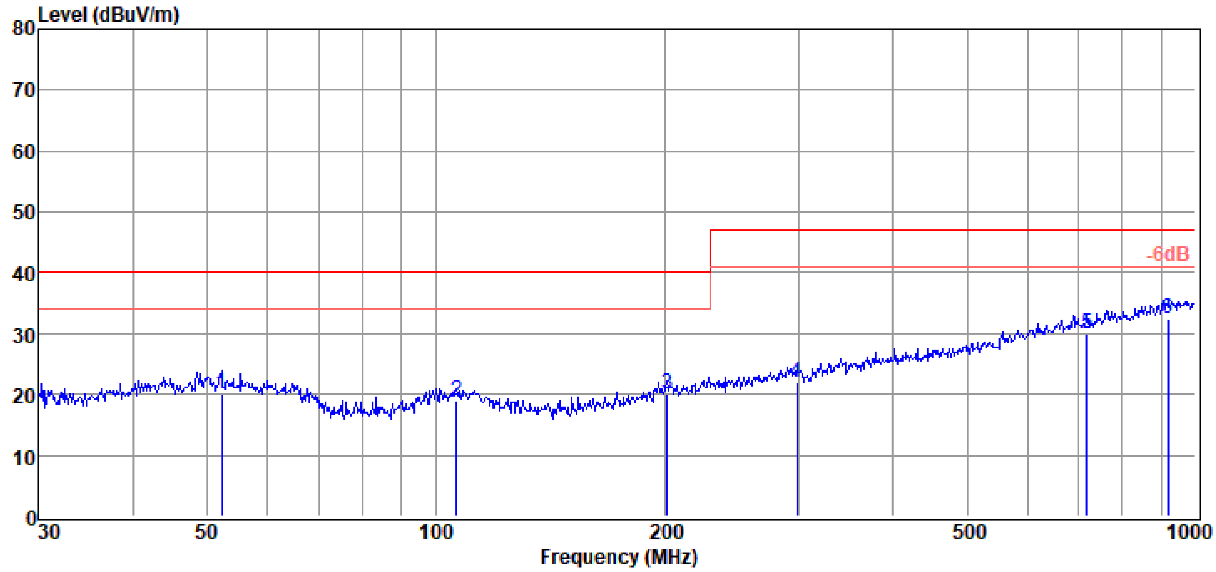
Test Mode : Working mode

Condition : Temp:23.5°C,Humi:65.4%

Antenna/Distance : 2025 9163 1#/3m/HORIZONTAL

Memo : S25091517-003

Data: 2



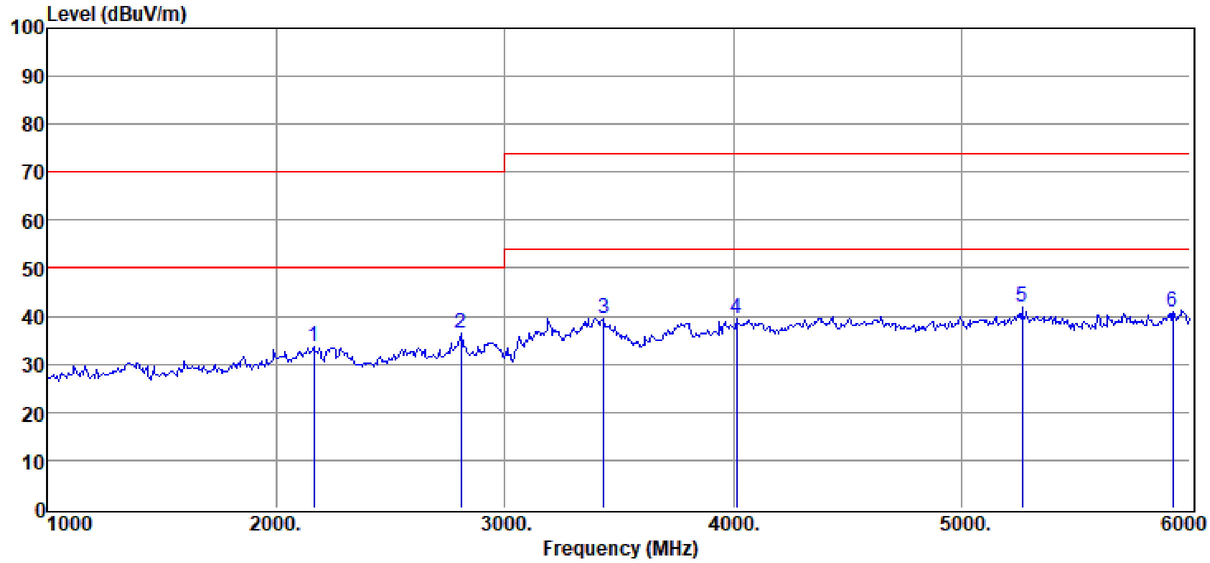
Item (Mark)	Freq. (MHz)	Read Level (dBµV)	Antenna Factor (dB/m)	Cable Loss dB	Result Level (dBµV/m)	Limit Line (dBµV/m)	Margin (dB)	Detector	Polarization
1	52.21	2.41	13.82	3.78	20.01	40.00	19.99	QP	HORIZONTAL
2	106.39	3.03	11.62	4.20	18.85	40.00	21.15	QP	HORIZONTAL
3	201.39	4.06	11.23	4.76	20.05	40.00	19.95	QP	HORIZONTAL
4	298.27	2.91	13.76	5.22	21.89	47.00	25.11	QP	HORIZONTAL
5	719.20	3.32	20.06	6.71	30.09	47.00	16.91	QP	HORIZONTAL
6	919.29	3.28	22.03	7.24	32.55	47.00	14.45	QP	HORIZONTAL

Note: 1. Result Level = Read Level + Antenna Factor + Cable loss.
2. If Peak Result complies with QP limit, QP Result is deemed to comply with QP limit.
3. Test setup: RBW: 120 kHz, VBW: 300 kHz, Sweep time: auto.

TR-4-E-009 Radiated Emission Test Result

Test Site	:	DDT 3m Chamber 1#	D:\2025 RE 1# Report data\Q25091517-2E\1203 RE-H.EM6		
Test Date	:	2025-12-03	**Tested By**	:	Li guilin
Power Supply	:	DC 3.3V from control board	**Test Mode**	:	Working mode
Condition	:	Temp:23.5°C,Humi:65.4%	**Antenna/Distance**	:	2025 9120D 1#/3m/ VERTICAL
Memo	:	S25091517-003			

Data: 1



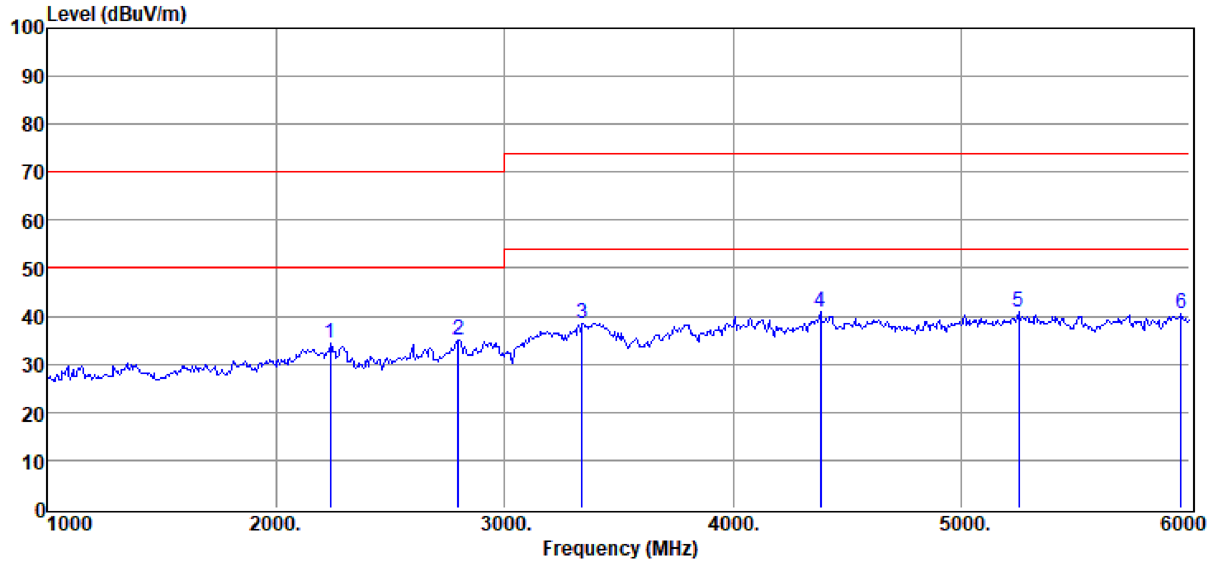
Item (Mark)	Freq. (MHz)	Read Level (dBµV)	Antenna Factor (dB/m)	PRM Factor dB	Cable Loss dB	Result Level (dBµV/m)	Limit Line (dBµV/m)	Margin (dB)	Detector	Polarization
1	2165.00	54.57	27.20	53.08	4.95	33.64	70.00	36.36	Peak	VERTICAL
2	2810.00	55.82	28.26	53.02	5.47	36.53	70.00	33.47	Peak	VERTICAL
3	3435.00	58.12	28.68	53.26	6.02	39.56	74.00	34.44	Peak	VERTICAL
4	4015.00	56.69	29.93	53.59	6.53	39.56	74.00	34.44	Peak	VERTICAL
5	5265.00	55.39	32.88	53.54	7.19	41.92	74.00	32.08	Peak	VERTICAL
6	5925.00	54.61	33.05	54.40	7.65	40.91	74.00	33.09	Peak	VERTICAL

Note: 1. Result Level = Read Level + Antenna Factor + Cable loss - PRM Factor.
2. If Peak Result complies with AV limit, AV Result is deemed to comply with AV limit.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.
4. According to standard requirements, the radio carrier and harmonic frequencies of the samples are not included in the test results.

TR-4-E-009 Radiated Emission Test Result

Test Site	:	DDT 3m Chamber 1#	D:\2025 RE 1# Report data\Q25091517-2E\1203 RE-H.EM6		
Test Date	:	2025-12-03	**Tested By**	:	Li guilin
Power Supply	:	DC 3.3V from control board	**Test Mode**	:	Working mode
Condition	:	Temp:23.5°C,Humi:65.4%	**Antenna/Distance**	:	2025 9120D 1#/3m/ HORIZONTAL
Memo	:	S25091517-003			

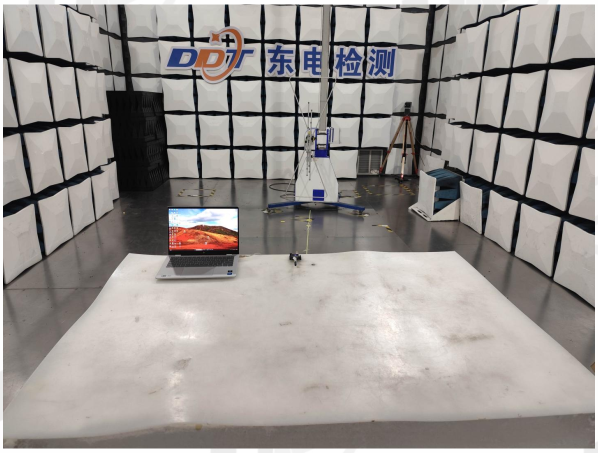
Data: 2

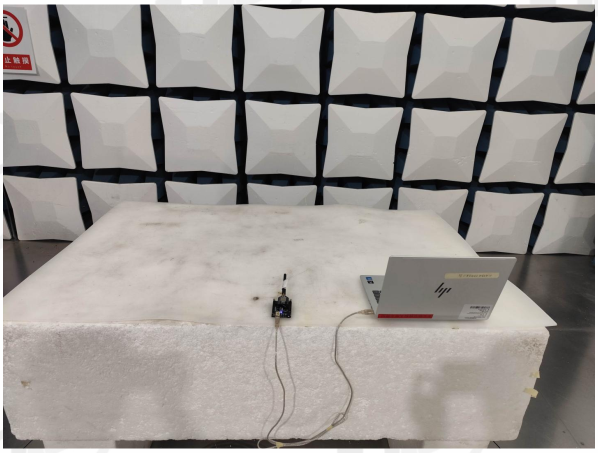


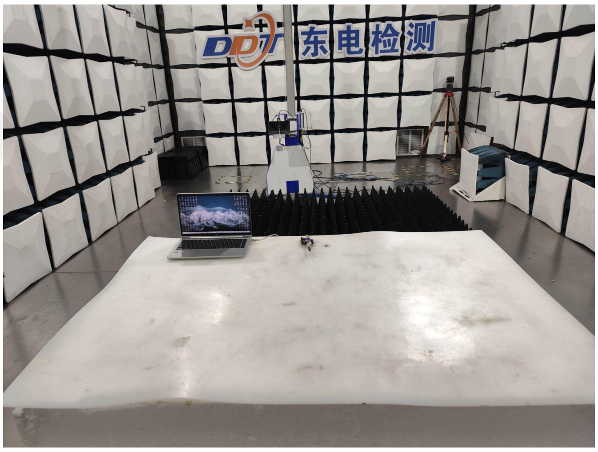
Item (Mark)	Freq. (MHz)	Read Level (dBμV)	Antenna Factor (dB/m)	PRM Factor dB	Cable Loss dB	Result Level (dBμV/m)	Limit Line (dBμV/m)	Margin (dB)	Detector	Polarization
1	2240.00	55.36	26.96	53.08	5.01	34.25	70.00	35.75	Peak	HORIZONTAL
2	2800.00	54.31	28.30	53.02	5.47	35.06	70.00	34.94	Peak	HORIZONTAL
3	3340.00	56.82	29.06	53.20	5.93	38.61	74.00	35.39	Peak	HORIZONTAL
4	4385.00	56.88	30.68	53.45	6.70	40.81	74.00	33.19	Peak	HORIZONTAL
5	5250.00	54.12	33.00	53.53	7.18	40.77	74.00	33.23	Peak	HORIZONTAL
6	5965.00	54.37	32.86	54.45	7.68	40.46	74.00	33.54	Peak	HORIZONTAL

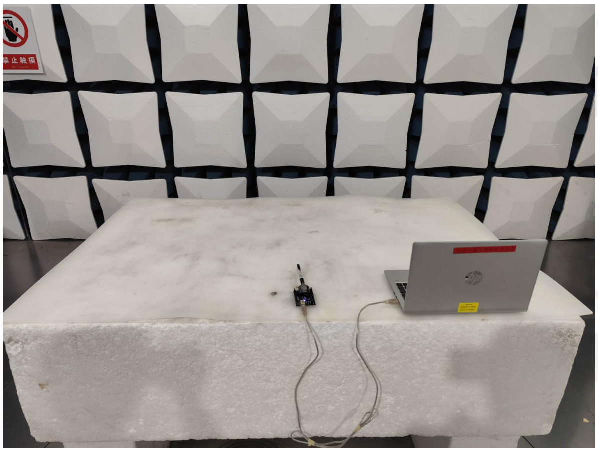
Note: 1. Result Level = Read Level + Antenna Factor + Cable loss - PRM Factor.
2. If Peak Result complies with AV limit, AV Result is deemed to comply with AV limit.
3. Test setup: RBW: 1 MHz, VBW: 3 MHz, Sweep time: auto.
4. According to standard requirements, the radio carrier and harmonic frequencies of the samples are not included in the test results.

3.8. Test photo







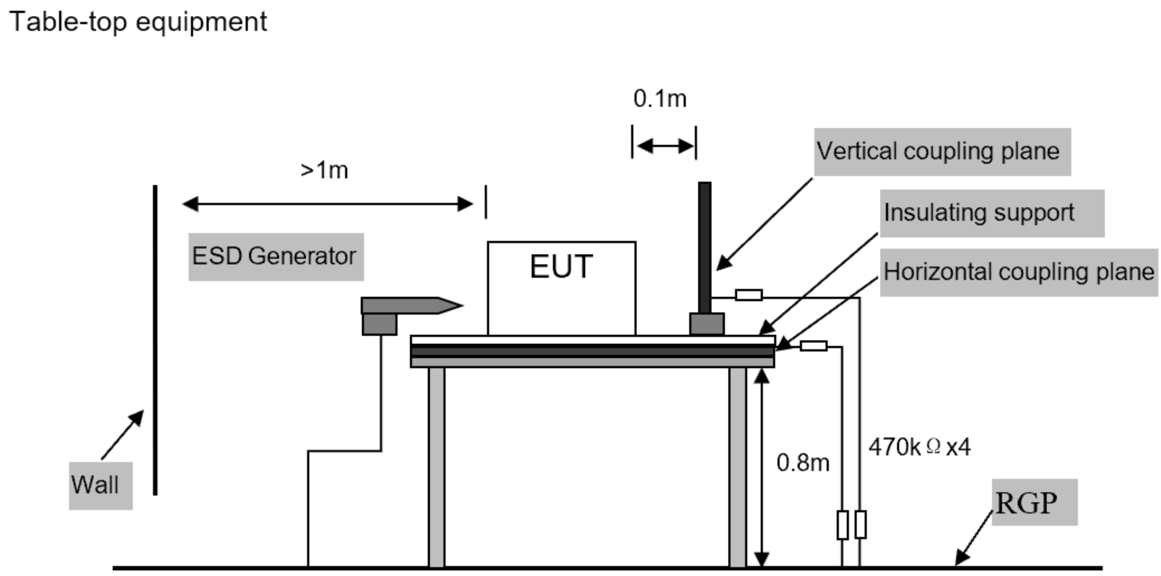


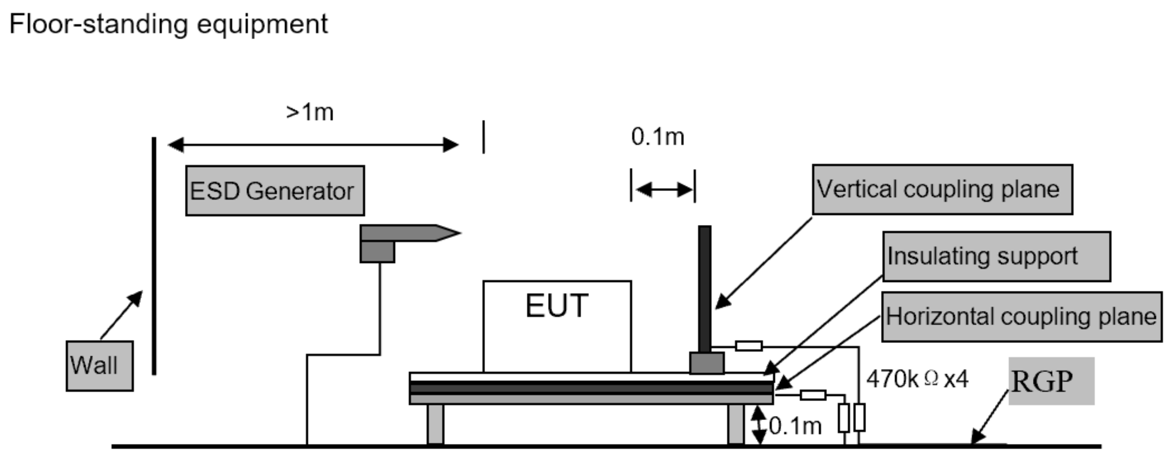
4. Electrostatic Discharge Test

4.1. Test equipment

Equipment	Manufacturer	Model No.	Equipment No.	Cal. Due to
ESD Generator	TESEQ AG	NSG 437	DDT-ZC00247	2026/08/15

4.2. Block diagram of test setup





4.3. Test levels and performance criterion

Test Level		Performance Criteria
Air Discharge	±2kV, ±4kV and ±8kV	B
Contact Discharge	±4kV	
Performance criteria B description: During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test. After the test, the EUT shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the EUT is used as intended.		

4.4. Assistant equipment used for test

Assistant equipment	Manufacturer	Model number	Description	other
Laptop	HP	HP ZHAN 66 Pro A 14 G3	5CD1014N1K	/

4.5. Test procedure

Air Discharge:

The test was applied on non-conductive surfaces of EUT. The round discharge tip of the discharge electrode was approached as fast as possible to touch the EUT. After each discharge, the discharge electrode was removed from the EUT. The generator was re-triggered for a new single discharge and repeated 20 times for each pre-selected test point. This procedure was repeated until all the air discharge completed.

Contact Discharge:

All the procedure was same as air discharge. Except that the generator was re-triggered for a new single discharge. The tip of the discharge electrode was touching the EUT before the discharge switch was operated.

Indirect discharge for horizontal coupling plane:

At least 20 single discharges were applied to the horizontal coupling plane, at points on each side of the EUT. The discharge electrode positions vertically at a distance of 0.1m from the EUT and with the discharge electrode touching the coupling plane.

Indirect discharge for vertical coupling plane:

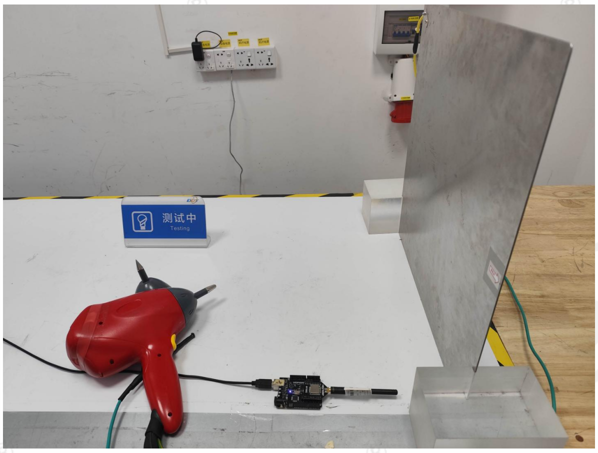
At least 20 single discharges were applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m X 0.5m, was placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges were applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

4.6. Test result

Test Site: 1# ESD Room	Test Date: 2025/12/03--2025/12/03
Condition: 23°C,43%RH,101kPa	Test Engineer: Elosky Liu
Memo: /	

EUT Name: LoRa module			EUT Model: RFM90C		
Sample No.: S25091517-003			Test Mode: Working mode		
Power supply: DC 3.3V from control board			Memo:/		
Measure parameters: 20 times at each point for contact discharge; 20 times at each point for air discharge. 1 second interval for each discharge.					
Type of discharge	Test Level	Test Point	Required	Observation	Result
Contact to EUT	±4kV	/	B	/	/
Contact to Coupling Planes	±4kV	Coupling Planes	B	A	Pass
Air	±2 kV /±4 kV /±8kV	/	B	/	/
Observation Description: A: Normal performance within limits specified by the manufacturer requestor or purchaser.					

4.7. Test photo

	/
	/

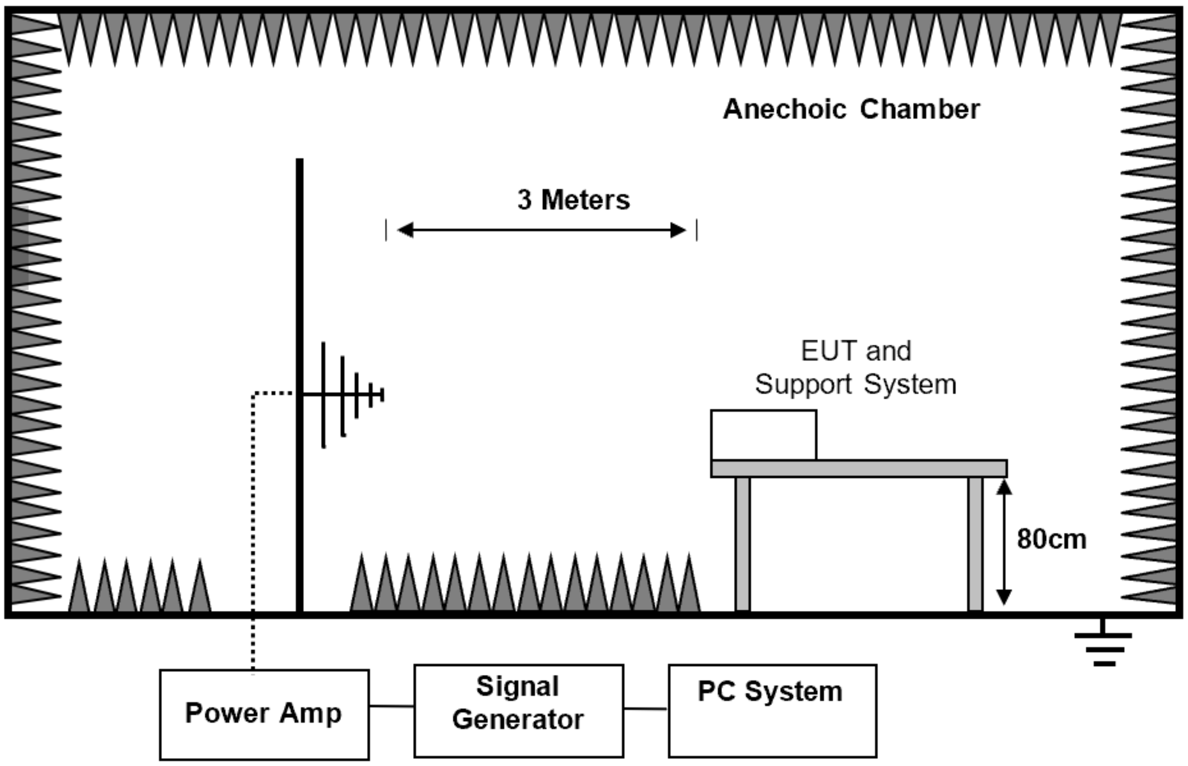
5. Continuous Radio Frequency Disturbances Test

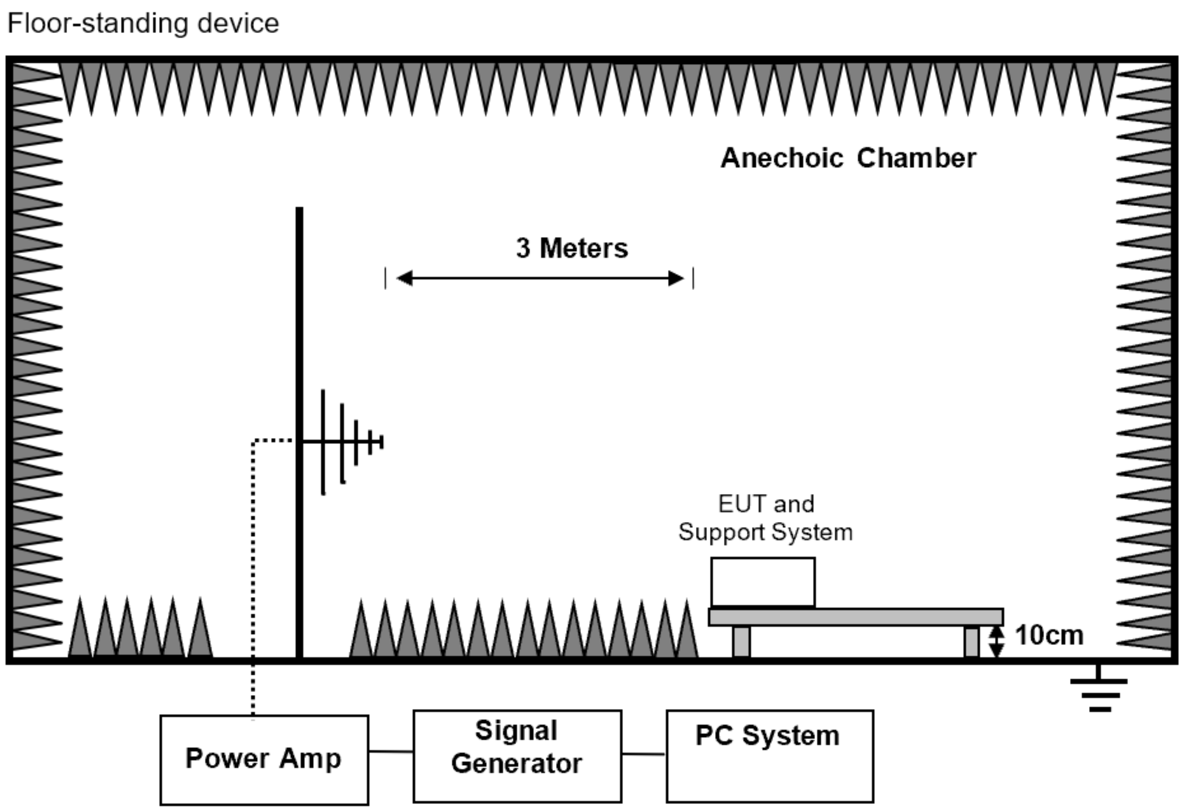
5.1. Test equipment

Equipment	Manufacturer	Model No.	Equipment No.	Cal. Due to
Power meter	Agilent	E4417A	DDT-ZC00934	2026/07/06
Power sensor	Agilent	E9323A	DDT-ZC00938	2026/08/10
Power sensor	Agilent	E9323A	DDT-ZC00939	2026/08/10
RS Test Software	SKET	EMC-S	DDT-ZC01455	/
Log-Periodic Antenna	schwarzbeck	STLP 9129	DDT-ZC04673	2026/10/19
Audio Analyzer	R&S	UPL	DDT-ZC04520	2026/10/19
Field probe	PMM	EP-601	DDT-ZC00933	2026/01/08
MXG Vector Signal Generator	Agilent	N5181A	DDT-ZC05321	2026/10/19
Power Amplifier	Micotop	MPA-80-1000-250	DDT-ZC05601	2026/10/19
Power Amplifier	Micotop	MPA-1000-6000-100	DDT-ZC05602	2026/10/19
Matrix Switch	SKET	LDSPDT80M6G	DDT-ZC05734	/

5.2. Block diagram of test setup

Table-top device





5.3. Test levels and performance criterion

Test Level		Performance Criteria
Frequency	80 MHz to 6 GHz	A
Field Strength	3V/m measured unmodulated	
Modulation	AM modulated to a depth of 80% by a sinusoidal audio signal of □1 kHz/□400 Hz	
Step Size	1% increments	
Dwell time	>0.5 Sec.	

5.4. Assistant equipment used for test

Assistant equipment	Manufacturer	Model number	Description	other
Laptop	HP	HP ZHAN 66 Pro A 14 G3	5CD1014N1K	/

5.5. Test procedure

The field sensor is placed on the EUT table (Ground clearance height reference "block diagram of test setup") which is 3 meters away from the transmitting antenna. Through the signal generator, power amplifier and transmitting antenna to produce a uniformity field strength around the EUT table from frequency range specified and records the signal generator ‘s output level at the same time for whole measured frequency range. Then, put EUT and its simulators on the EUT turn table and keep them 3 meters away from the transmitting antenna which is mounted on an antenna tower and fixes at 1 meter height above the ground. Using the recorded signal generator’s output level to measure the EUT from frequency range specified and both horizontal & vertical polarization of antenna must be set and measured. Each of the four sides of EUT must be faced this transmitting antenna and measures individually.

5.6. Test result

Test Site: 4# 1m Chamber	Test Date: 2025/12/05--2025/12/05
Condition: 22°C,48%RH	Test Engineer: Chen Jintang
Memo: /	

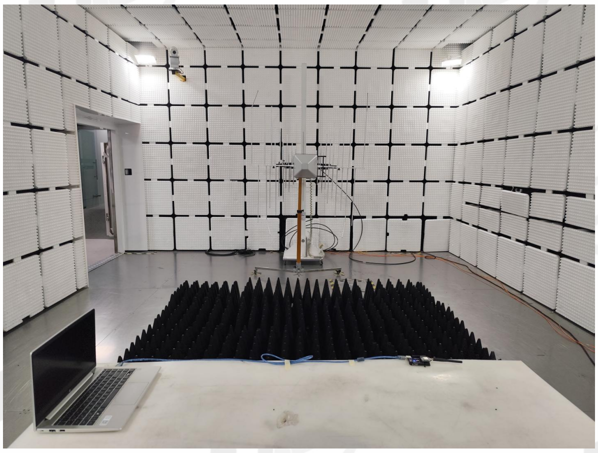
EUT Name: LoRa module	EUT Model: RFM90C
Sample No.: S25091517-003	Test Mode: Working mode
Power supply: DC 3.3V from control board	Memo:/

Steps: 1%, Dwell time 1s, 1kHz 80% AM modulation

Test frequency	Level	EUT Position	Antenna Polarization	Required	Observation	Result
80MHz to 6000MHz	3V/m	Front	H	A	A	Pass
			V	A	A	Pass
		Left	H	A	A	Pass
			V	A	A	Pass
		Rear	H	A	A	Pass
			V	A	A	Pass
		Right	H	A	A	Pass
			V	A	A	Pass

Observation Description:
A: Normal performance within limits specified by the manufacturer requestor or purchaser.

5.7. Test photo

	/
/	/

6. Photos of the EUT

Please refer to DDT-Q25091517-2E appendix I.

-----------End Report----------